

DOVER AFB, DE, USA

WMO: 724088

Lat: 39.133N

Lon: 75.467W

Elev: 9

StdP: 101.22

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13707

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-7.8	-18.1	0.8	-8.3	-15.9	0.9	-6.3	12.7	3.5	11.8	1.8	4.8	310	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	33.5	24.5	32.2	24.0	30.8	23.6	26.6	30.4	25.7	29.5	25.0	28.7	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.6	20.9	28.0	24.6	19.7	27.3	23.9	18.8	26.8	82.7	30.3	78.8	29.7	75.7	28.7	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.6	8.5	-12.8	35.9	2.8	1.7	-14.8	37.1	-16.4	38.1	-18.0	39.0	-20.0	40.2		
(5)			-13.6	27.5	2.6	1.0	-15.5	28.3	-17.0	28.9	-18.5	29.4	-20.4	30.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	1.3	2.7	6.7	12.4	17.6	22.6	25.1	24.3	20.8	14.7	8.6	4.3		
	DBStd	9.39	5.77	5.33	5.16	4.73	4.28	3.52	2.77	2.67	3.47	4.38	4.52	5.00		
	HDD10.0	919	278	211	130	25	1	0	0	0	0	8	78	188		
	HDD18.3	2463	529	437	362	188	67	7	0	0	15	131	292	434		
	CDD10.0	2188	7	7	27	98	236	378	469	442	323	153	37	12		
	CDD18.3	691	0	0	1	10	44	134	211	184	88	17	1	0		
	CDH23.3	5646	0	0	10	87	364	1116	1945	1483	546	92	2	1		
	CDH26.7	1941	0	0	2	18	109	386	768	497	145	16	0	0		
(14) Wind	WSAvg	4.0	4.5	4.5	4.8	4.6	3.8	3.6	3.3	3.2	3.5	3.8	4.0	4.2		
(15) Precipitation	PrecAvg	1147	89	76	104	89	101	100	106	115	113	83	82	91		
	PrecMax	1599	195	178	235	206	198	252	362	409	301	229	166	213		
	PrecMin	562	9	16	6	12	15	18	27	30	7	0	14	20		
	PrecStd	229	42	41	48	41	48	54	63	69	67	52	46	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	20.8	24.4	29.1	32.0	34.4	35.8	34.7	32.9	28.4	22.9	20.6		
		MCWB	16.4	16.0	16.3	20.0	23.5	24.8	25.1	25.5	24.0	21.8	18.3	17.7		
	2%	DB	16.5	17.2	20.8	26.1	29.8	32.4	33.9	32.6	30.3	25.9	20.2	17.4		
		MCWB	14.6	13.1	14.0	18.7	22.2	23.7	24.6	24.3	23.3	20.5	16.5	15.4		
	5%	DB	13.0	13.7	17.8	23.1	27.4	30.7	32.4	31.1	28.0	23.6	17.9	14.9		
		MCWB	10.6	10.1	13.1	17.0	21.0	23.3	24.3	23.8	22.3	19.3	14.9	12.7		
	10%	DB	9.8	10.6	15.0	20.6	25.1	28.9	30.9	29.4	26.3	21.7	15.9	12.2		
		MCWB	7.6	7.4	11.1	15.7	19.8	22.7	23.9	23.4	21.4	18.2	13.2	10.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.5	17.5	18.2	22.9	25.1	26.8	28.2	28.0	26.0	24.0	19.8	18.8		
		MCDB	18.7	20.2	22.4	24.6	29.0	30.0	31.1	31.0	29.3	26.0	21.1	19.4		
	2%	WB	14.8	14.1	16.1	19.9	23.5	25.5	26.8	26.1	24.9	22.1	17.8	15.9		
		MCDB	16.4	15.8	19.4	24.3	27.7	29.8	30.8	30.2	27.9	24.5	19.4	17.1		
	5%	WB	11.3	10.7	14.0	18.3	22.3	24.6	25.8	25.2	23.9	20.6	15.8	13.1		
		MCDB	12.8	13.0	16.7	22.0	26.1	28.8	29.8	28.9	26.5	22.5	17.3	14.3		
	10%	WB	7.8	7.8	11.7	16.6	21.0	23.7	25.1	24.4	22.9	19.0	13.9	10.4		
		MCDB	8.9	9.9	14.9	20.1	24.0	27.5	28.9	27.7	25.4	21.0	15.4	11.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.2	9.0	9.7	10.3	9.6	9.4	9.3	9.0	9.1	9.6	9.2	8.3		
		MCDBR	12.3	13.2	14.0	13.6	12.3	11.4	10.9	10.5	10.6	11.0	11.6	11.0		
		MCWBR	10.4	10.2	9.4	7.8	6.4	5.0	4.1	4.2	5.1	6.7	8.4	9.8		
	5% WB	MCDBR	11.8	12.8	12.9	11.7	10.7	10.0	9.3	8.9	8.8	9.1	9.8	10.4		
		MCWBR	10.7	11.1	10.2	8.1	6.4	5.2	4.4	4.4	4.9	6.9	8.6	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.320	0.368	0.398	0.444	0.499	0.516	0.509	0.421	0.362	0.328	0.312		
	taud		2.466	2.447	2.353	2.298	2.200	2.093	2.060	2.065	2.286	2.456	2.504	2.514		
	Ebn at Noon		859	905	887	879	841	793	774	768	821	844	836	836		
	Edh at Noon		80	93	113	128	144	160	164	159	120	92	77	72		
(44) All-Sky Solar Radiation	RadAvg		2.06	2.93	3.88	4.99	5.61	6.13	6.08	5.39	4.42	3.26	2.35	1.81		
	RadStd		0.18	0.32	0.32	0.38	0.66	0.46	0.28	0.34	0.46	0.34	0.20	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.54	N/A	N/A	N/A	+0.61	+0.81	N/A	N/A	+145	+91
(47) Regional (69 neighbors)		+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+79

Nomenclature: See separate page